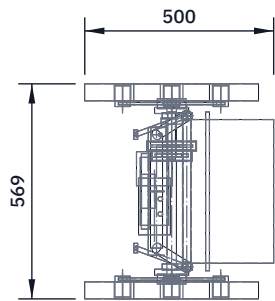


CONCEPT, NOT FOR FABRICATION

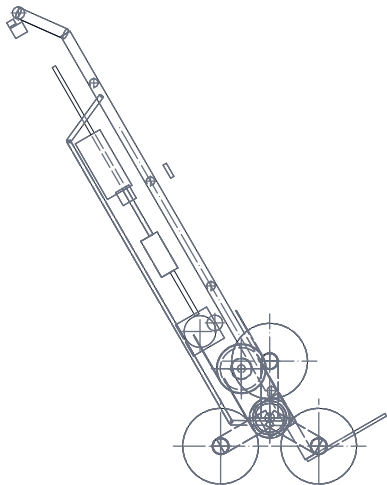
StepClimber · [github.com/BoujeeEnjinia1701/stepclimber](https://github.com/BoujeeEnjinia1701/stepclimber)

| REV | DESCRIPTION                                               | DATE       | BY |
|-----|-----------------------------------------------------------|------------|----|
| P1  | Preliminary general arrangement, TRL 3                    | 2026-09-25 | AC |
| P2  | 200 mm wheels, 150 mm arms, two-stage drive (SCM-DDR-002) | 2026-09-25 | AC |
| P3  | Layout and labels tidied                                  | 2026-09-30 | AC |



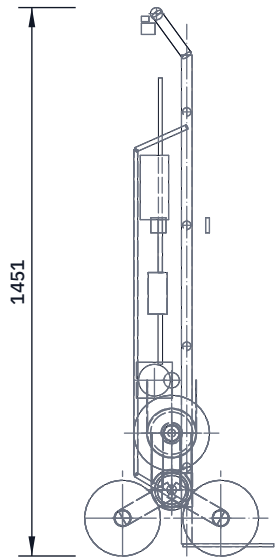
TOP VIEW

Scale 1:20; looking down (along -Z)



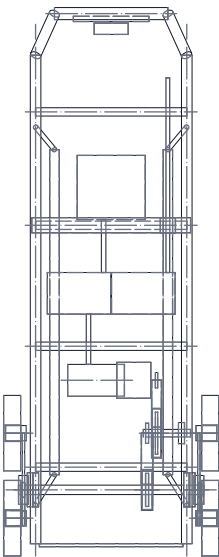
FRONT VIEW, CLIMBING POSE

Tilted 30 deg about the shaft; scale 1:20; looking at the front (along +Y)



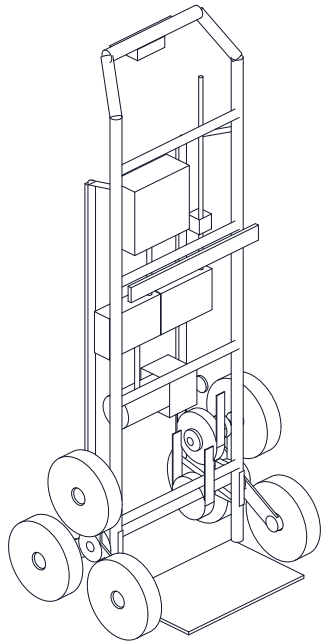
FRONT VIEW

Scale 1:20; looking at the front (along +Y)



RIGHT VIEW

Scale 1:20; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS AND INTERFACES (MM)

Overall width 569 (wheel outer faces); upright height 1451

Clusters: arm 150, wheel dia 200, spacing 259.8

Shaft 25 dia at 175.0 above floor; bearings 400 + 28 apart

Cluster mid-planes 524 apart; rails 400 apart, 28 tube

Drive 06B 10T:35T then 08B 10T:20T (7:1)

Countershaft 150 and gearmotor 290 above the shaft, along the frame

Shaft sprocket OD 88; guard radius 53.9 (limit 54.2, R15 met)

Grip 1260 from the shaft along the frame; toe plate 240 x 380

Pack cradle 75 x 180 x 170, center 800 above shaft

Climbing tilt 30 deg back; tilt window +/-6 deg; 60 kg rated stair load

STEPCLIMBER

[github.com/BoujeeEnjinia1701/stepclimber](https://github.com/BoujeeEnjinia1701/stepclimber)

General arrangement

|                                                                                                                  |             |                                                                                                     |                    |                 |                                             |
|------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------|--------------------|-----------------|---------------------------------------------|
| DRAWING NO.<br>SCM-DWG-001                                                                                       |             |                                                                                                     | REV<br>P3          | SHEET<br>1 of 1 | SIZE<br>ANSI B                              |
| SCALE<br>1:20                                                                                                    | UNITS<br>mm | PROJECTION<br> | DATE<br>2026-09-30 |                 | DRAWN<br>Amish Chadha                       |
| MATERIAL / NOTES<br>Steel frame and spiders; solid rubber wheels; see bom/bom.csv. Figures from SCM-CAL-001 v0.2 |             |                                                                                                     |                    |                 |                                             |
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